MVP Software Development

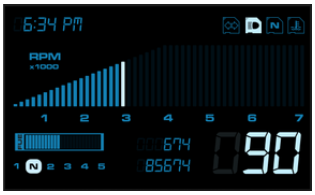
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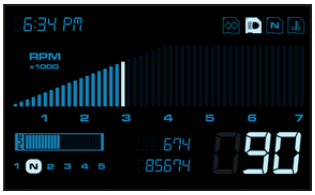
General Requirements

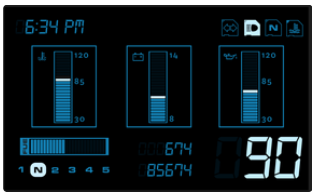
Sub Topic	Acceptance Criteria	Business Requirement	Comments / Screenshots

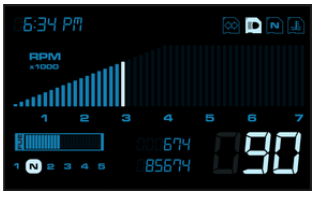
General	• Device readouts **will not** come from CANBUS reading from the bike • Device readouts **will** come directly via wiring from sensors on the bike	**Must Have**	
Device Control	• Must be controllable by a bar-mounted control • Control must have 4 Axis pad and a central button • Button function will be: ◦ D-UP: Directional Pad Up ◦ D-DOWN: Directional Pad Down ◦ D-LEFT: Directional Pad Left ◦ D-RIGHT: Directional Pad Right ◦ SELECT: Centre Button	**Must Have**	

Device UI	Device is Broken down into two main components **Console** • Used when riding • Contains Speedometer, Tachometer, Gear Selector, Odometer/Tripmeters, Voltmeter, Oil Pressure, Engine Temperature, Alerts **Configurator** • Only accessible when bike is stationary (speedometer reads zero) • Contains following screens: ◦ Performance ◦ Read Outs ◦ Settings	**Must Have**	

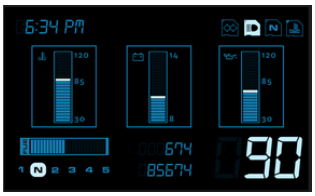
			 BARONESS 12:34 AM PERFORMANCE READ OUTS SETTINGS EXIT SPEEDOMETER MAX SPEED --- TACHOMETER MAX RPM --- UNIT OF MEASUREMENT KM/H MPH

Console UI	**UI Layout** • The following will always show on screen: ◦ Speedometer ◦ Alerts ◦ Odometer ◦ Tripmeter ◦ Digital Clock • The following will always show on the console if activated in the Configurator: ◦ Fuel Gauge ◦ Gear Selector **View Selection** • The following will show on screen as part of the UI, that the user can horizontally scroll through: ◦ View A: ▪ Tachometer ◦ View B: ▪ Engine Temperature ▪ Volt Meter ▪ Oil Pressure • User can cycle between View A and View B using D-LEFT and D-RIGHT • If user continues to press D-Left or D-Right it will continue to cycle between View A and View B	**Must Have**	

			View A  View B

Console: View A	View A presents: • Tachometer AND • Speedometer • Alerts • Odometer • Tripmeter • Digital Clock • Fuel Gauge (If activated in Configurator) • Gear Selector (If activated in Configurator)	**Must Have**	

			View A

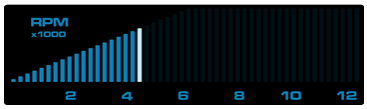
Console: View B	View B presents: • Engine Temperature (If activated in Configurator) • Volt Meter (If activated in Configurator) • Oil Pressure (If activated in Configurator) AND • Speedometer • Alerts • Odometer • Tripmeter • Digital Clock • Fuel Gauge (If activated in Configurator) • Gear Selector (If activated in Configurator)	**Must Have**	

			View B
Deactivated Features	• Certain features must always be active (i.e. Speedometer) • Other features can choose to be activated/deactivated in the configurator Where the following features are deactivated, the following should occur to the: **Fuel Gauge** • Gauge is hidden from view in the console • Gear selector will move up to where the Fuel Gauge usually is in the UI • All other items in the UI will remain static **Gear Selector** • Gear Selector is hidden from view in the console • All other items in the UI will remain static ℹ Where both Fuel Gauge and Gear Selector are deactivated, all other features will remain static and the space where they would be will be replaced with a graphic **[Graphic TBC]** **Engine Temperature** • Engine Temperature is hidden from view in the console **Volt Meter** • Volt Meter is hidden from view in the console **Oil Pressure** • Oil Pressure is hidden from view in the console ℹ **Engine Temperature | Volt Meter | Oil Pressure Gauge** Where 1 of 3 is deactivated: • Other 2 will group together • The 2 will re-center in the middle of the screen Where 2 of 3 is deactivated: • The remaining 1 will re-center in the middle of the screen Where 3 of 3 are deactivated • Screen B **will not** be accessible via the controls	**Must Have**	

	• If user presses D-LEFT or D-RIGHT it will stay on the same screen		

Console Requirements

Sub Topic	Acceptance Criteria	Business Requirement	Comments / Screenshots

Speedometer	• Speed will be calculated using a sensor attached to the front wheel • Speed must be accurately represented on the screen ◦ No lag can occur between sensor readings / backend calculations and the representation in the UI • Presented as numbers on the screen up to 3-digits • Speed must be calculable and presentable in either: ◦ mph = Miles per hour ◦ km/h = Kilometres per hour • User can select readout to display as either km/h or mph in the Configurator • Speedometer Colour Scheme: ◦ Active Digits: [HUE], 100, 90, 100% ◦ Inactive Digits: [HUE], 100, 37, 12% **Speed Sensor** • Device will be sold with the Speed Sensor • Users will need to: ◦ Attach the sensor to the bikes front wheel ◦ Manually input the distance the sensor is placed from the centre of the wheel	**Must Have**	 Configurator screen - Sensor Placement & Units of Measurement selection **Speed calculation example** **Speedometer-H… sor.zip** 07 Jan 2024, 09:48 PM
Tachometer	• UI presents Tachometer as: ◦ Bar that is filled from left to right ◦ Numbers under the tachometer represent current revolutions in the 000's ▪ Only numbers 1-2 digits should be presented • 1 = 1000 RPM, 12 = 12,000 RPM ▪ the highest RPM value that can be presented is 20 ▪ Where a max value is > 10: • Max value Even: numbers will start at 2 and increase in increments of 2 (2, 4, 6, 8, 10) • Max value Odd: numbers will start at 1 and increase in increments of 2 (1, 3, 5, 7, 9, 11) ◦ Tachometer Bars: ▪ There should always be 50 bars shown in the UI ▪ Each bar represents a component of the RPM range • Example: ◦ Rev range = 0 - 12,000 RPM ◦ Bar Value = 12000 / 50 = 240 ◦ Bar 1 range = 0 - 240 RPM ◦ Bar 6 range = 1200 - 1440 RPM ◦ Bar 38 range = 8880 - 9120 RPM ▪ When a bike revs through the rev range the following animation should occur:	**Must Have**	 Example: Tachometer movement  Example: Tachometer **Tachometer Sensor Reading** Different bikes provide a different number ignition impulses per crankshaft revolution. This will be configurable by the User in the Configurator, and they will be able to choose Option 1 (0.5) or Option 2 (1)

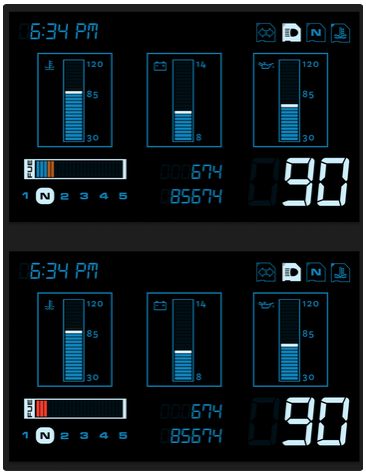
	- Bars below the current RPM: [HUE], 100, 37, 100% - Bar within the current RPM: [HUE], 100, 90, 100% - Bars above the current RPM: [HUE], 100, 37, 12% - Tachometer must be accurately represented on the screen - No lag can occur between sensor readings / backend calculations and the representation in the UI - User can set Maximum RPM - Up to 2 digits can be entered - Screen UI will show 000 after digits so users understand the number they've entered represents in the 000's - Examples: 6 entered = 6000 Max RPM 12 entered = 12000 Max RPM - Maximum safe RPM must be able to be set		If a vehicle is equipped with multiple ignition coils, only the impulses from the coil with the signal cable attached (i. e. the one actually measured) are counted. **Examples:** - Single cylinder, 4-stroke, one ignition coil = 0.5 - Example Bikes: SR 500, XT 500 - Single cylinder, 4-stroke, two ignition coils = 1 - Example Bikes: GSXR 1000 - Two cylinders, 4-stroke, one ignition coil = 1 - Example Bikes: Harley Davidson, dual fire - Two cylinders, 4-stroke, two ignition coils = 0.5 - Example Bikes: Harley Davidson, single fire 99% of all motorcycles use either 0.5 or 1. If user is unsure which to use it is recommended that they choose Option 1, exit the setup menu and start the engine. If only half the idle rpm value is displayed, enter the value 0.5.

Odometer / Tripmeter	• User should be able to read the distance the bike has driven: ◦ Odometer: ▪ Total distance the vehicle has ridden since installation of the device ▪ Cannot be reset by the user ◦ Tripmeter: ▪ Distance the vehicle has driven since the user has set the Trip to zero ▪ User should be able to reset the trip meter to zero by: • Using D-UP / D-DOWN to highlight the Tripmeter • Holding down OK button until UI show trip at 0.00 ▪ User should be able to switch between Trip 1 and Trip 2: • This can be done by: ◦ Using D-UP / D-DOWN to highlight the Tripmeter ◦ Using D-LEFT / D-RIGHT to switch between the Trips • User should be able to read this in: ◦ Miles per hour ◦ Kilometres per hour ◦ If a user switches from Mph to Km/h in the configurator, then this will update in both Trip and Odometer • Odometer numerical reading should be between: ◦ 0 ◦ 99999.99 • Tripmeter numerical reading should be between: ◦ 0 ◦ 99999.99 • Trip & Odometer readings should be saved when bike is switched off	**Must Have**	
Digital Clock	• Time able to be configured in		
Fuel Gauge	• Users Fuel Range can be measured via two different methods: ◦ Readout from in-built sensor on bike ◦ Manually set so visual representation of fuel is based off a range of kilometres **Sensor Readout** • User will wire the device directly to the sensor in the fuel tank of the bike	**Should Have**	**Example: Fuel Sensor** GAUGE, FUEL  Fuel Sensor

- Fuel Gauge sensor will advise level of fuel in the tank
- Gauge will be full when all bars are [HUE], 100, 37, 100%
- Gauge bars will change to [HUE], 100, 37, 12% from right to left as fuel depletes
- Fuel Warning of Amber and Red will show as fuel reaches 10-15%

Manual KM Range

- Users can set a fuel range in KM's or Miles via the Configurator
- This will then be represented in the Fuel Gauge in the console
- Fuel Gauge will show as 25 Bars
  - First 20 Represented in [HUE]
  - Next 2 in Amber, representing early low fuel warning
  - Last 3 in Red, representing the need to fill up immediately
- KM range input in the configurator will be be divided as follows:
  - Non- Reserve Range = [Range Entered]
  - Fuel Bar Value = [Range Entered] / 20
  - Reserve Range = [Fuel Bar Vaue] * 5
  - Total Range = [Reserve Range] + [Non-Reserve Range]
- When Fuel is full, all bars are [HUE], 100, 37, 100%
- As bike travels, bars will change from [HUE], 100, 37, 100% to [HUE], 100, 37, 12%
  - Bars will change starting from the right, and change every [Fuel Bar Value] km's
- User can reset the fuel gauge back to full by:
  - Using D-UP / D-DOWN to highlight the fuel gauge
  - Holding down OK button until UI updates



Warning States

Gearing	• Gearing should be able to be hidden (where gear readings aren't available for the bike) • Where gearings are available, they will show as: ◦ Default: 1,N,2,3,4,5,6 • Users will be able to change the gear structure of their bike	**Should Have**	
Alerts	• User must be able to determine if Highbeam is switched on/off: ◦ When highbeams are switched on, the highbeam alert will be illuminated on the screen ◦ When highbeams are switch off, the highbeam alert will **not** be illuminated on the screen • Neutral Gear: Must advise when the motorcycle is in Neutral ◦ When the bike is in Neutral, the Neutral indicator will be alert on the screen ◦ When the bike isn't in Neutral, the Neutral indicator will **not** be alert on the screen • Engine Temperature: Must switch on when sensor advised engine is overheating ◦ When the bike receives a high temperature reading, the Engine Temperature alert will be illuminated on the screen ◦ When the bike isn't receiving a high temperature reading, the Engine Temperature alert will **not** be illuminated on the screen ◦ Engine Temperature alert will switch on when Engine Temperature sensor reads greater than 104 Degrees • Indicators: Must switch on when indicators are on ◦ When the bike's indicator switch is pressed, the indicator alert will be illuminated on the screen ◦ When the bike's indicator switch isn't pressed, the indicator alert will **not** be illuminated on the screen ◦ The bike's alert will flash on and off at the same pace as the indicator flashes on the vehicle	**Must Have**	
Engine Temperature	• User can choose to switch on Engine Temperature readings from the bikes engine via the configurator • If a user switches this on then they will be able to view the reading via the Console: View B • Engine Temperature range should be: 82°C to 104°C: • Warning sign will be as follows: ◦ **TBC**: Bars Blue ◦ **TBC:** Bars Orange ◦ **TBC**: Bars Red	**Should Have**	
Volts	• User can choose to switch on volt readings from the bikes battery via the configurator	**Should Have**	

	• If a user switches this on then they will be able to view the reading via the Console: View B • Volt gauge range should be: ◦ 12V: 8V - 16V ◦ 6V: TBC • Warning sign will be as follows: ◦ **>= 12.6V**: Bars Blue ◦ **<12.60V & > 11.31V:** Bars Orange ◦ **<= 11.31V**: Bars Red		
Oil Pressure	• User can choose to switch on Oil Pressure readings from an installed Oil Pressure sensor • Baroness will sell a 3rd party sensor for users to install, and configuration of readings should be based off this sensor • If a user switches this on then they will be able to view the reading via the Console: View B • Oil Pressure gauge can be set as either BAR or PSI in the Configurator • Based on this choice, Gauge range will show as: ◦ BAR: 0 - 8 ◦ PSI: 0 - 116 • Warning sign will be as follows: ◦ **TBC**: Bars Blue ◦ **TBC:** Bars Orange ◦ **TBC**: Bars Red	**Should Have**	

Configurator Requirements

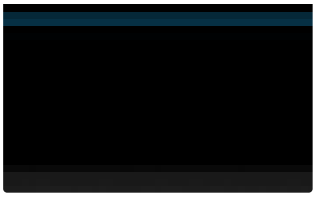
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Performance Screen	The Performance Screen allows users to update the following: **Speedometer Placement** Allows users to advise: • Distance from the central axis of the front wheel that users have placed the hall sensor. • This allows Speedometer readings to be accurate **Tachometer** Allows users to advise: • Maximum range for the Tachometer in RPM • Safe Range for the Tachometer in RPM **Units of measurement** • Km/h vs Mph • Centigrade vs Farenheit	**Must Have**	
Read Outs Screen	The Readouts screen allows users to configure: • Gearbox • Volt Meter • Fuel Gauge **Gearbox** • User is able to activate/deactivate Gear Selector by checking/unchecking the checkbox next to the 'Gear Selector' title ◦ See Activate/Deactivate requirement for UI changes • User can determine: ◦ The gear range (i.e 1,2,3,4) ◦ Location of Neutral (i.e. 1,N,2,3,4 or N,1,2,3,4,5) **Motorcycle Gearbox Information** • Motorcycles move through gears via clicking either up or down via a gear selector next to the riders foot • Most motorcycles Select Neutral between 1st and 2nd gear, however some bikes have Neutral before 1st so users need to be able to select it's location. **Volt Metre** • User is able to activate/deactivate Volt Metre by checking/unchecking the checkbox next to the 'Battery Health' title ◦ See Activate/Deactivate requirement for UI changes • Users need to be able to select between either a bike that runs a 6V or 12V battery system ◦ Switching between 6V and 12V will update Voltage and Warning ranges in the Console: View B UI for	**Must Have**	

either battery type

Fuel Gauge

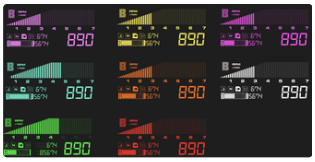
- User is able to activate/deactivate Fuel Gauge by checking/unchecking the checkbox next to the ‘Fuel Gauge’ title
  - See Activate/Deactivate requirement for UI changes
- User can select between a Sensor or Manual fuel calculation
- Where a user selects Manual Sensor:
  - Range Input box will appear and field will be activate
  - User Warning will pop up on the screen
    - Warning can only be closed if user selects ‘I Accept’
    - Once pressed, user will be returned to the Read Outs screen
  - User will be able to enter up to 4 digits with no decimal places
  - If user has selected KM’s / Mph on Performance Screen then user can enter either in the box
- For Functionality read Fuel Gauge requirements







Settings Screen	The Settings screen allows users to configure: • Colour Scheme • Language • Device ID **Colour Scheme** • User can update the overall UI's colour scheme by selecting from one of six options: ◦ Blue ◦ Red ◦ Indigo ◦ Green ◦ Amber ◦ White • When a user steps through the list of colours on the screen, the UI will change to the selected UI colour in the moment it's selected • For Confirmation of Colour Scheme variables and codes read the below ◦ Colour Scheme **Language** • User can select between the following languages in their UI: ◦ English ◦ Spanish ◦ Hindi ◦ Japanese ◦ Mandarin • Device will default to English when first turned on • When a user steps through the list of languages on the screen, the UI will change to the selected language in the moment it's selected **Device ID** • Each device will be given a Unique ID • The Unique ID will be presented on the Settings screen	**Must Have**	

			 Example: Console: View A with different Colour Schemes

Questions

1	Question	Answer
2		
3		
4		
5		
6		